Weekly Update, 9–13 Nov 2026

Brian Zhou · CSE 5999 hiking-speed project · for Danni Beaulieu · 13 Nov 2026

In short: I removed the Bay Area and fixed two small bugs, so I'm rebuilding the data. The new numbers come this weekend. I also found that terrain needs the real ground height, not super-sharp maps.

1. I removed the Bay Area

It was the orange dot on the map: the OSM server kept giving up on it, so its data came only from the lab's copy. I made the download ask for smaller boxes, and that worked. But after 20 hours only a third was done, and 96% of the data came from two accounts that log car drives, which we throw away anyway. So I removed it. It was only 6% of our hikes. Now all 16 regions get their data the same way.

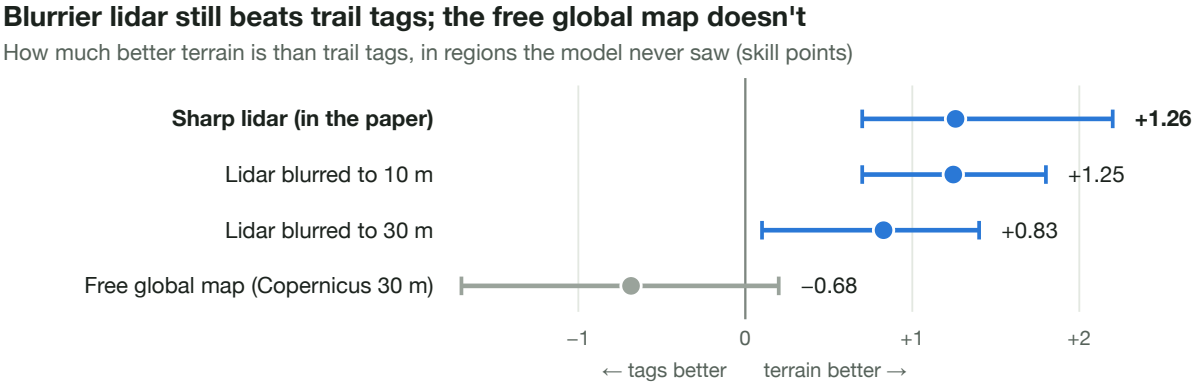
2. I fixed two small bugs

- In four areas, the OSM server gave up partway, so some older hikes were missing. I downloaded them again.
- Some anonymous hikes that crossed the edge of a download box were cut into pieces and counted more than once. Now the pieces are joined back together.

So I'm rebuilding all 16 regions. Some counts drop a little (White Mountains: 218 hikes become 210), but the trail data the models use changed by less than 2% in the six regions I compared.

3. How sharp does the elevation map need to be?

Our terrain features use very sharp lidar maps. I blurred them on purpose and also tried the free global map:



Dots are averages; lines show the 95% range. Measured on the old data (17 regions); I'll rerun it on the new data.

Blurred to 10 m, nothing is lost. Blurred to 30 m, terrain still beats the trail tags. The free global map (Copernicus, made by radar) does not: in forests it often sees the treetops instead of the ground. **So terrain needs the real ground height, not super-sharp maps.**

4. Other news

- **Jonathan's answer:** the lab has older GPS files (2008–2016), from a dataset OSM published itself, and nothing after 2022. I'll ask Jonathan if they still have times. If they do, we could add more hikes to every region.
- **The paper:** the IGARSS 2027 template is out, and our draft already fits it (4 pages plus references). It's due 10 Jan; I plan to submit around 8 Jan.

Questions for you

1. Is it OK that I removed the Bay Area, so the paper has 16 regions?
2. I can't travel outside the US. Will you or Tianyi be in Iceland in July? If not, who could present for me?
3. If Jonathan's older files have times, should I add them after the 20 Nov draft? (They're on lab storage my account can't reach yet.)

Next: new numbers this weekend. Mon–Thu I rewrite the draft; on Fri 20 Nov it goes to you, Tianyi and Prof. Jacobs.